

Sec 5.5 Nonhomogeneous Eqn (part 1)

Recall: ① $y'' + p(x)y' + q(x)y = f(x)$ non-homog. if $f \neq 0$ (f is not constant 0)

associated homog eqn $y'' + p(x)y' + q(x)y = 0$

General solution for ① is

$$y = \boxed{C_1 y_1 + C_2 y_2} + \boxed{y_p}$$

"y_c"

solution to assoc. homog

eqn ("complementary function")

"particular solution
(for ①) to account for $f \neq 0$ "

so gen solution for nonhomog. linear ODE

is $\boxed{y = y_c + y_p}$ (similar for 3rd, 4th, ... order)

Example for method:

Find general sol for $y'' - 4y = 3x$ ②

$$y = y_c + y_p$$

Step 1 Compute y_c ($y'' - 4y = 3x^2$)

assoc. homog eq. is $y'' - 4y = 0$

ch. eq is $r^2 - 4 = 0$

$$\Rightarrow r = \pm 2$$

Gen sol for. assoc. homog eqn is

$$\underline{y_c = c_1 e^{-2x} + c_2 e^{2x}}$$

Step 2 Compute/find y_p , using a "trial function"

(this is the "method of undetermined coefficients")

Idea:

- need y_p to satisfy $\underline{y_p'' - 4y_p = 3x^2}$
- want to have $y_p, y_p', y_p'', \dots$ all add/subtract/mix in $(y_p'' - 4y_p)$ to result in $\underline{3x^2}$
- If we "trial"/"guess" $y_p = e^{kx}$ (k TBD), then
 $y_p = e^{kx}, y_p' = k e^{kx}, y_p'' = k^2 e^{kx}$; all these contain e^{kx} . Thus, adding/mixing $y_p'' - 4y_p$ will probably still have e^{kx} , and is unlikely to result in $3x^2$

- It is better / best to pick something for y_p to have same "form" as $f(x) = 3x^2$.

$$f(x) = 3x^2 \quad (\text{degree 2 polynomial} \quad 3x^2 + 0x + 0)$$

so let's trial $y_p = \underline{A} + \underline{B}x + \underline{C}x^2$ (2nd-deg. poly)

"undetermined
coeffs"

$$y_p = A + Bx + Cx^2$$

$$y_p' = B + 2Cx$$

$$y_p'' = 2C$$

$$\Rightarrow y_p'' - 4y_p = 3x^2$$

$$2C - 4(A + Bx + Cx^2) = 3x^2 + 0x + 0$$

$$\underline{-4Cx^2} - \underline{4Bx} + (\underline{2C-4A}) = \underline{3x^2} + \underline{0x} + \underline{0}$$

$$\Rightarrow \left. \begin{array}{l} -4C = 3 \\ -4B = 0 \\ 2C - 4A = 0 \end{array} \right\} \Rightarrow \begin{array}{l} \underline{C = -3/4} \\ \underline{B = 0} \\ 4A = -6/4 \Rightarrow \underline{A = -3/8} \end{array}$$

thus, $\boxed{y_p = -\frac{3}{8} - \frac{3}{4}x^2}$ is a solution for $y'' - 4y = 3x^2$

Step 3: Full gen. sol for $\rightarrow$ is

$$\underline{y} = y_c + y_p = \left[c_1 e^{-2x} + c_2 e^{2x} + \left(-\frac{3}{8}\right) - \frac{3}{4}x^2 \right]$$

Note: Why include y_c ?

with $y_p = -\frac{3}{8} - \frac{3}{4}x^2$, we know

$$y_p'' - 4y_p = 3x^2$$

• y_c , by defin, satisfies $y_c'' - 4y_c = 0$

• So then $y_c + y_p$ a solution? $y'' - 4y = 3x^2$

$$(y_c + y_p)'' - 4(y_c + y_p) = 3x^2$$

$$y_c'' + y_p'' - 4y_c - 4y_p = 3x^2$$

$$\underbrace{(y_c'' - 4y_c)}_0 + \underbrace{(y_p'' - 4y_p)}_{3x^2} = 3x^2$$

• Need to include y_c to make complete/all solution(s)

Ex: $y'' - y' - 2y = 2e^{3x}$

step 1: compute y_c , solution for $y'' - y' - 2y = 0$

$$\dots \underline{y_c = c_1 e^{-x} + c_2 e^{2x}}$$

step 2: trial y_p

$$f(x) = 2e^{3x} = c e^{3x}$$

$$\text{try } y_p = Ae^{3x}$$

$$\begin{array}{l} y_p = Ae^{3x} \\ y_p' = 3Ae^{3x} \\ y_p'' = 9Ae^{3x} \end{array}$$

Want $y_p'' - y_p' - 2y_p = 2e^{3x}$ ($y'' - y' - 2y = \underline{2e^{3x}}$)

$$9Ae^{3x} - 3Ae^{3x} - 2Ae^{3x} = 2e^{3x}$$

$$\underline{4A}e^{3x} = 2e^{3x}$$

$$4A = 2 \Rightarrow A = \frac{1}{2}$$

$$\boxed{y_p = \frac{1}{2}e^{3x}}$$

Step 3 Combine, gen sol $y = \boxed{y_c} + \boxed{y_p}$ ↗

$$\boxed{y = \boxed{c_1 e^{-x} + c_2 e^{2x}} + \boxed{\frac{1}{2}e^{3x}}}$$

• $f(x)$ has x^k (incl. x^0 , a constant) $\Rightarrow$ include $x^k, x^{k-1}, \dots, x, x^0 (=1)$ in trial y_p .

• $f(x)$ has e^{kx} , then include " Ae^{kx} " in y_p

• if $f(x)$ has $\cos(kx)$ or $\sin(kx)$ $\Rightarrow$ include BOTH $A\cos(kx)$ and $B\sin(kx)$ in y_p

Ex: $y'' - y' - 2y = \cos x$

Step 1 y_c : gen sol for $y'' - y' - 2y = 0$
same as last example

$$\underline{y_c = c_1 e^{-x} + c_2 e^{2x}}$$

Step 2 $f(x) = \cos(1x)$, so trial

$$y_p = \underline{A \cos x + B \sin x}$$

$$y_p' = -A \sin x + B \cos x$$

$$y_p'' = -A \cos x - B \sin x$$

plug into $y_p'' - y_p' - 2y_p = \cos x$

$$\begin{aligned} (-A \cos x - B \sin x) - (-A \sin x + B \cos x) \\ - 2(A \cos x + B \sin x) \end{aligned}$$

$$(-3A - B) \cos x + (A - 3B) \sin x = \cos x$$

$$\left. \begin{aligned} -3A - B &= 1 \\ A - 3B &= 0 \end{aligned} \right\} \Rightarrow B = -\frac{1}{10}, A = -\frac{3}{10}$$

so $\boxed{y_p = -\frac{3}{10} \cos x - \frac{1}{10} \sin x}$

is a solution (particular)
for $y'' - y' - 2y = \cos x$

Step 3 Gen solution
for $y'' - y' - 2y = \cos x$ is $y = y_c + y_p = c_1 e^{-x} + \dots$

Ex: $y'' - y' - 2y = \underbrace{x}_{f(x)} \underbrace{e^{3x}}$

deg 1 polynomial $(1x+0)$ exponential e^{3x}

trial function $y_p = (A+Bx)e^{3x}$

Ex: $y'' - y' - 2y = \underline{x \sin(2x)}$

trial function y_p

$$= (A+Bx)(C \cos(2x) + D \sin(2x))$$

$$\cancel{AC} \boxed{\cos(2x)} + \cancel{BC} \boxed{x \cos(2x)} + \cancel{AD} \boxed{\sin(2x)} + \cancel{BD} \boxed{x \sin(2x)}$$

really let's use

$$y_p = A \cos(2x) + Bx \cos(2x) + C \sin(2x) + Dx \sin(2x)$$